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ANNEXURE A
DATA INTERPRETATION – ANSWERS

1: Hough Transform & Shape Detection

Scenario: Detect linear features (roads, rivers) in satellite images.

A: High-resolution grayscale satellite images

Answer: The input consists of high-resolution grayscale satellite images of urban and rural areas, approximately 1024 × 1024 pixels, stored in TIFF or PNG format. These images provide the intensity information needed to identify linear structures.

Interpretation: Roads, rivers and canals can appear as elongated regions or edges. Grayscale representation makes edge detection and Hough Transform processing simpler.

B: Noise, clouds, shadows and partial occlusions

Answer: The images may contain noise caused by clouds, shadows, vegetation and sensor effects. Some portions of roads or rivers may also be hidden.

Interpretation: Noise should be reduced using Gaussian or median filtering before applying edge detection. The Hough Transform can still detect a line when part of it is missing because the visible edge points continue to vote for the same line.

C: Multiple linear features with intersections

Answer: The images can contain several roads, rivers and canals, and these features may intersect.

Interpretation: The Hough Transform can detect multiple lines because each possible line receives votes in parameter space. Different orientations produce different peaks in the accumulator.

D: Optional synthetic dataset

Answer: A synthetic dataset can contain lines drawn at random angles and lengths. The known line positions and angles can be used as ground truth.

Interpretation: Synthetic images provide a controlled way to test whether the Hough Transform correctly detects line position, length and orientation.

Answer Summary for Question 1

- Preprocess the image using noise filtering.
- Apply Canny edge detection.
- Apply the Hough Line Transform using $\rho = x \cos \theta + y \sin \theta$.
- Find peaks in the Hough accumulator.

- Convert detected parameters into line segments.
- Evaluate using precision, recall, F1-score and line-position/orientation error.

2: PCA & Shape Priors for Object Recognition

Scenario: Segment organs in MRI scans using shape priors.

A: MRI slices

Answer: The input consists of grayscale MRI slices of approximately 512 × 512 pixels showing organs such as the liver or heart.

Interpretation: MRI intensity information helps identify organ regions, but boundaries may be weak or affected by noise.

B: Training set with annotated organ masks

Answer: A training set of 50–100 annotated organ masks is used to construct a PCA-based statistical shape model.

Interpretation: The masks provide the ground-truth shape of the organ. They are aligned and converted into numerical shape vectors before PCA is applied.

C: Test set with partial boundaries or noise

Answer: The test set contains 10–20 MRI images where organ boundaries may be incomplete or noisy.

Interpretation: The learned shape prior helps estimate missing boundaries and prevents segmentation from following irrelevant noisy structures.

D: File formats

Answer: The MRI data can be stored in DICOM or PNG format.

Interpretation: DICOM is commonly used for medical imaging and can contain imaging metadata, while PNG provides a simpler image representation for processing.

Answer Summary for Question 2

- Align training organ shapes.
- Calculate the mean shape.
- Perform PCA and obtain eigenvectors/eigenvalues.
- Select principal components representing major shape variations.
- Represent a new shape as $x \approx \bar{x} + Pb$.
- Use the shape prior with image information for segmentation.
- Evaluate using Dice score, IoU and Hausdorff distance.

3: Machine Learning Methods (SVM, HMM, GMM, EM)

Scenario: Human activity recognition from video feature trajectories.

A: Video sequences

Answer: The input consists of 30–60 FPS video sequences, approximately 640 × 480 pixels, containing activities such as walking, running and jumping.

Interpretation: The sequence of frames provides information about how the human body changes over time.

B: Feature trajectories

Answer: Features can be joint positions represented by x and y coordinates or optical-flow vectors extracted for each frame.

Interpretation: Joint coordinates describe body movement, while optical flow represents the direction and magnitude of image motion.

C: Training labels

Answer: Each training video is assigned an activity label such as walking, running or jumping.

Interpretation: Labels enable supervised learning methods such as SVM to learn the relationship between extracted motion features and activities.

D: Optional synthetic CSV data

Answer: A CSV file containing simulated joint coordinates over time can be used as synthetic training or testing data.

Interpretation: Synthetic data is useful for validating algorithms before using real video data.

Answer Summary for Question 3

- SVM learns a classification boundary between activity classes.
- HMM models the temporal sequence of human movements using hidden states and transitions.
- GMM models feature distributions as a mixture of Gaussian components.
- EM estimates GMM parameters through repeated E-step and M-step operations.
- Evaluate with accuracy, precision, recall, F1-score and a confusion matrix.

4: Deep Learning (CNNs, Transformers, GANs)

Scenario: Detect rare obstacles in urban self-driving car images.

A: RGB camera images

Answer: The input consists of RGB urban street images of approximately 1024 × 768 pixels in JPG or PNG format.

Interpretation: RGB information provides color and appearance features that help identify objects in road scenes.

B: Labeled dataset with bounding boxes

Answer: Each rare obstacle, such as a traffic cone or animal, is labelled using a bounding box and object class.

Interpretation: Bounding boxes tell the detector where the obstacle is located and what type of object it is.

C: Small dataset of 500–1000 images

Answer: The real dataset contains approximately 500–1000 labelled images of obstacles.

Interpretation: Because the dataset is relatively small, transfer learning, augmentation and class-balanced sampling should be used to reduce overfitting and improve rare-object detection.

D: Optional GAN-generated images

Answer: GANs can generate synthetic images or obstacle examples to augment the real dataset.

Interpretation: Synthetic samples can increase diversity and help address the rare-object problem. However, generated images should be checked for realism and correct labels.

Answer Summary for Question 4

- CNNs learn hierarchical spatial features for object detection.
- Transformers use attention to capture relationships between image regions and global scene context.
- GANs can generate additional synthetic samples for data augmentation.
- Use transfer learning because the real dataset is small.
- Evaluate using IoU, precision, recall, mAP and inference speed.

5: Graph-Based Learning for Shape Correspondence

Scenario: Find correspondences between 3D human face meshes.

A: 3D face meshes

Answer: The input consists of 3D human face meshes in OBJ or PLY format, with approximately 5,000–50,000 vertices.

Interpretation: Each face is represented as a collection of connected 3D vertices and polygonal faces. The mesh structure preserves the geometry of the face.

B: Training set with annotated landmark correspondences

Answer: The training set contains 20–50 face meshes with annotated landmark correspondences.

Interpretation: Landmarks such as the nose tip, eye corners, mouth corners and chin provide known matching locations for training the correspondence model.

C: Test set

Answer: The test set contains 5–10 unseen face meshes.

Interpretation: The unseen meshes are used to determine whether the trained model can generalize correspondence to new faces.

D: Graph adjacency matrices

Answer: Each mesh can be converted into a graph $G = (V, E)$, where vertices are nodes and mesh connections are graph edges. An adjacency matrix records the connectivity between vertices.

Interpretation: Graph representation preserves local facial structure and is suitable for graph-based learning methods such as Graph Neural Networks.

Answer Summary for Question 5

- Construct a graph from mesh connectivity.
- Extract vertex features such as coordinates, normals and curvature.
- Train using annotated landmark correspondences.
- Learn vertex embeddings using graph-based learning.
- Match vertices using embedding similarity and geometric constraints.

- Do not rely on vertex indices because different meshes can have different vertex counts.
- Evaluate using landmark localization error, PCK, correspondence accuracy and geodesic error.

Overall Conclusion

The five questions apply different computer vision and machine learning techniques to real-world data. The Hough Transform is appropriate for detecting linear structures in satellite images. PCA provides a statistical shape prior for MRI organ segmentation. SVM, HMM, GMM and EM provide different approaches for human activity recognition from temporal features. CNNs and transformers are suitable for rare-obstacle detection, while GANs can support data augmentation. Graph-based learning is appropriate for correspondence between irregular 3D face meshes because it preserves mesh connectivity and local geometry.